

What we're doing

Does continuing the quick #67 contacts-v1 1.5B run (protein-contacts-1_5b-3.5e-4-contacts-v1-unmasked-3b5cf2`) for **another epoch** with a re-heated learning rate (a cosine warm-restart from the final `step-11999` checkpoint) lower eval loss — and does it improve contact recapitulation on the exp82 benchmark?

Why

The #67 run was a single un-tuned cosine decay over ~ 2.7 epochs (final eval/loss ****2.980****, bpb 0.4232). Its LR decayed to its floor, so the last steps made little progress. A ****warm restart**** — reload the weights, re-heat the LR, and run ~ 1 more epoch of shuffled data — should squeeze out additional loss reduction "for free" (no new data, ~ 1 epoch of compute) and give a second contacts-v1 checkpoint to compare against both #67 and the #61/#75 tuned sweep. Whether the loss gain translates into better contact prediction is the open question (exp82 showed #67 is near-chance at **de novo** contact prediction).

Results so far

Recipe + code complete (v5p-32, batch 512, re-heat LR $4.0e-4$, ~1125 steps, warm-start from #67 step-11999). The 2026-06-17 attempts were blocked before step 0 by a marin/levanter cache-reader mismatch on the iris TPU worker: `Sharded cache ledger missing input\_ids/0 count` while the ledger key is `input\_ids`. The branch now avoids that path by reading exp67's reused token caches as prebuilt `input\_ids` arrays (`PrebuiltLmDatasetFormat`) and disabling automatic cache rebuilds. Local direct loads of the real train and validation GCS caches both derive `input\_ids` and pass offset construction, so the next step is relaunching the v5p-32 warm restart and checking that it logs `train/loss`.